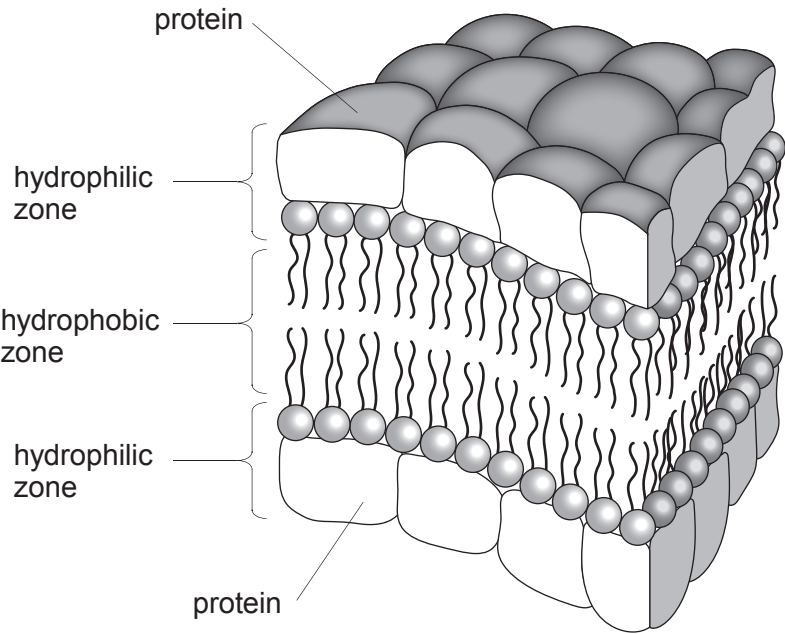


5. In 1935, Hugh Davson and James Danielli proposed a model of the arrangement of molecules in the plasma membrane of a cell.



- (a) Compare Davson-Danielli's model shown in the diagram to the currently accepted fluid mosaic model of the plasma membrane. [4]

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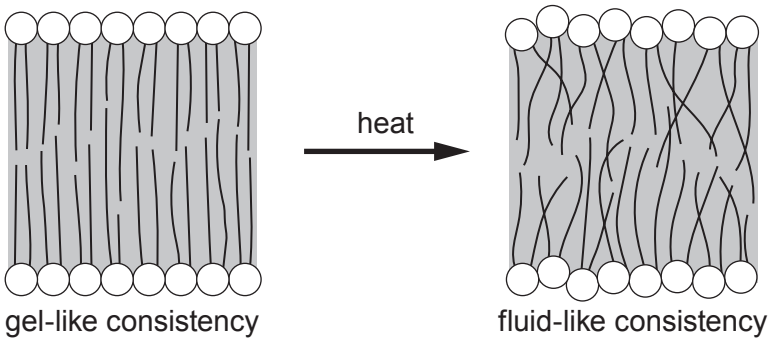
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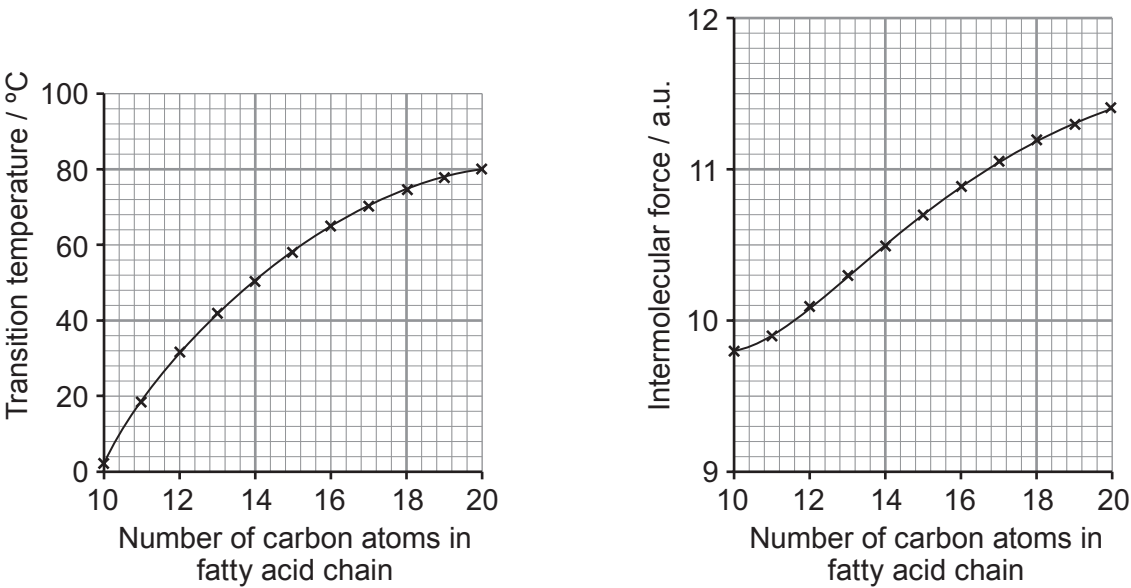
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Fluidity of the plasma membrane is due to the lateral movement of phospholipids. Increased fluidity occurs when the phospholipid bilayer changes from a more fixed gel-like consistency to a fluid-like consistency. Heating the molecules can result in increased fluidity.



The transition temperature, is the temperature at which the phospholipid bilayer changes from a gel like consistency to a fluid like consistency. This can also be affected by the length of the fatty acid chains which affect the strength of the intermolecular forces. These relationships are shown in the graphs below.



Examiner  
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- (b) (i) With reference to the graphs, describe the relationship between fatty acid chain length and the temperature at which the phospholipid bilayer becomes fluid. Explain your answer. [3]

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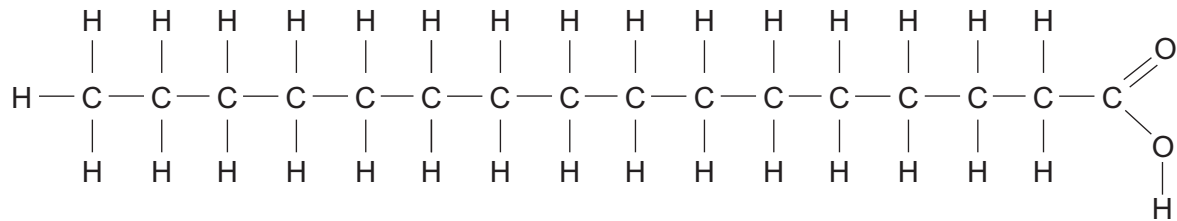
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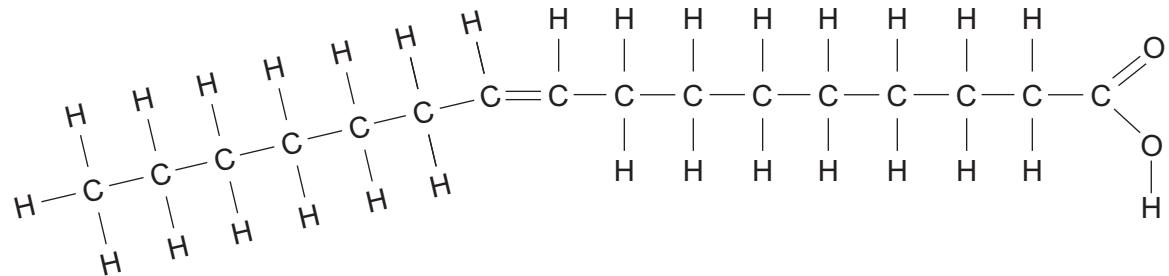


The packing of hydrocarbons also plays a significant role in affecting fluidity.  
Two examples of fatty acids that can be found in phospholipids are shown below.

Saturated Fatty Acid



Unsaturated Fatty Acid



- (ii) Suggest why the presence of unsaturated fatty acids in the phospholipid would increase fluidity of the membrane. [4]

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Cholesterol is a non-polar substance that is produced in the body and obtained from foods that come from animals (particularly egg yolks, meat, poultry, fish, and dairy products). The body needs this substance to build cell membranes. However, too much cholesterol increases a person’s risk of developing heart disease.

(iii) Suggest the position of cholesterol molecules within the plasma membrane. [1]

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(iv) State how a diet high in animal products and cholesterol can lead to a person developing heart disease. [2]

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